INCH-POUND

MIL-C-83729A
AMENDMENT 2
30 June 1994
SUPERSEDING
AMENDMENT 1
19 April 1977

MILITARY SPECIFICATION

CHOPPERS, ELECTROMECHANICAL
GENERAL SPECIFICATION FOR

This amendment forms a part of MIL-C-83729A, dated 1 February 1972, and is approved for use by all Departments and Agencies of the Department of Defense.

PAGE 2

* Standards, Military, delete "MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes."

3.2, line 2: Delete "(see 4.6)" and substitute "(see 4.5)".

PAGE 3

3.4.2.1: Delete title in its entirety and substitute "Dissimilar metals (not applicable to hermetically sealed choppers)".

PAGE 5

3.12, line 2: Delete "be evidence" and substitute "be no evidence".

PAGE 7

4.5, line 1: Delete "(see 3.3)" and substitute "(see 3.2)".

PAGE 8

* 4.7.1.2.1, Delete and substitute:

"4.7.1.2.1 Sampling plan. A sample of parts shall be randomly selected in accordance with table IIIa. If one or more defects are found, the lot shall be rescreened and the defective parts removed. After screening and removal of defective parts, a new sample of parts shall be randomly selected in accordance with table IIIa. If one or more defects are found in the second sample, the lot shall be rejected and shall not be supplied to this specification."

* Table III, delete and substitute:

" TABLE III. Group A inspection.

Inspection	Requirement paragraph	Test method paragraph
Seal	3.6	4.8.2
Dielectric withstanding voltage	3.8	4.8.4.1
Electrical characteristics 1/	3.7.1, 3.7.5, and 3.7.7	4.8.3.1.2, 4.8.3.5, and 4.8.3.7
Electrical characteristics 1/	3.7.2, 3.7.3, and 3.7.4	4.8.3.2, 4.8.3.3, and 4.8.3.4
Insulation resistance	3.9	4.8.5
Visual and mechanical inspection	3.1, 3.4, 3.5, 3.20, and 3.21	4.8.1

1/ With nominal drive only. "

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* 4.7.1.2.2, delete in its entirety.

* 4.7.1.2.3, renumber as 4.7.1.2.2.

* Add the following new table IIIa:

" TABLE IIIa. Group A inspection sampling plan.

Lot size	Sample size for tests in accordance with 3.6, 3.7.1, 3.7.5, and 3.7.7, 3.8	Sample size for tests in accordance with 3.7.2, 3.7.3, 3.7.4, 3.9, and visual and mechanical inspection
2 to 8	All	All
9 to 15	All	13
16 to 25	20	13
26 to 50	20	13
51 to 90	20	13
91 to 150	20	13
151 to 280	20	20
281 to 500	47	29
501 to 1,200	47	34
1,201 to 3,200	53	42
3,201 to 10,000	68	50

"

4.8.3.1.2, delete in its entirety and substitute:

"4.8.3.1.2 Dynamic (method II). Contact resistance measurement shall be made with the chopper in the circuit of figure 1.

a. Connect the chopper as shown on figure 1. Energize the coil with nominal rated voltage and frequency.

b. In the contact circuit, set the oscillator frequency between 20 kHz and 30 kHz.

c. Set R_1 from 50,000 ohms to value required (see 3.1 and 6.1.2) for maximum contact current rating (S_1 enclosed).

d. Set R_2 for 3 ohms. Set the gain of pre-amp and scope to present an average vertical deflection of 8 centimeters.

e. Open S_1, maintain a constant vertical deflection and adjust R_2 until V drop across contact is 1/5 to 1/10 R_2 V drop.

f. The dynamic contact resistance is the calculated ratio of the contact V drop times the resistance R_2 (Note: Lead length must be deducted from results.)."

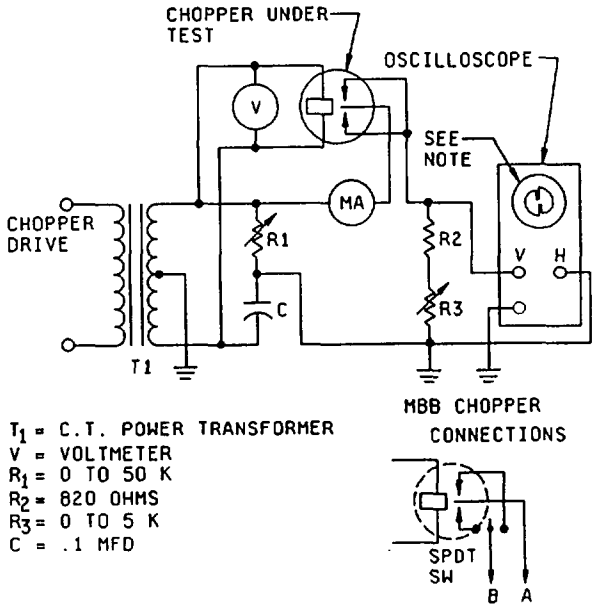
4.8.3.7, line 4: Following "oscilloscope", add "with a minimum diameter of 4 inches".

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FIGURE 2, delete and substitute the following:

"

BBM CHOPPER CONNECTIONS



NOTE: Waveform display, (see figure 3) shall be a minimum of 4.00 (101.6 mm) inches in diameter.

FIGURE 2. Circuit for measurement of waveform characteristics.

"

FIGURE 4: In upper right-hand corner, delete "EXITATION" and substitute "EXCITATION".

The margins of this amendment are marked with an asterisk to indicate where changes (additions, modifications, corrections, deletions) from the previous amendment were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

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